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PI Name:	Garofalo, Roberto M.D.		
Project Title:	Effect of Space Flight on Innate Immunity to Respiratory Viral Infections		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Human Research Program Risks:	None		
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Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	77555-0366	**Congressional District:**	14
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Task Description:	Our overarching hypothesis is that a combination of several factors from microgravity to stress, affect the human immune system during space flight, resulting in an impaired antiviral response in the respiratory tract. Three specific hypotheses that will be tested in this proposal are: 1. Space flight or ground conditions such as antiorthostatic suspension leads to an altered host response against two of the most common human respiratory pathogens, Respiratory Syncytial Virus (RSV) and Human metapneumovirus (hMPV); 2. This “deficient” antiviral response is characterized by a defect in both innate immunity, including IFN type I production in the respiratory tract, and T cell-mediated immune response; 3. The central events that cause this defect in antiviral immunity are the impaired trafficking and function of pDC in the respiratory mucosa.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space Applications: Space shuttle Discovery will bring NASA scientists one step closer to helping astronauts and the public discover ways to battle and prevent serious illness and infection. These investigations are expected to generate new scientific evidence of the immune pathways that are affected in antiviral host response during space flight and possible preventive or therapeutic approaches applicable to future space missions. Earth Applications: Understanding the function of the immune system during space flight may have great relevance to our understanding of the process of aging and/or stress-related immune-modulation (adjustments in the level of an immune response) on Earth. In particular, studying the innate host response against pathogens during/after space flight will provide novel data on the function of the respiratory mucosal response to viral pathogens. The goal of the Mouse Immunology-2 experiment is to discover what triggers and leads to an increased susceptibility to an infection. These findings can be used to help treat and prevent future astronauts from getting sick, as well as protect people with more vulnerable immune systems here on Earth, such as the elderly or young children.
	Respiratory Syncytial Virus (RSV): Human respiratory syncytial virus (RSV) is the leading cause of severe lower respiratory tract infection in infants and a major cause of morbidity and mortality in elderly and immunocompromised adults. Approximately two-thirds of infants are infected with RSV during the first year of life, and 90% have been infected one or more times by 2 years of age. However, both infants and adults are susceptible to recurrent infections. In the United States, RSV infection is the most common cause of respiratory disease leading to hospitalization in children, costing an estimated $300–$400 million annually. In the USA alone, RSV is responsible for 70 000 to 126 000 infant hospitalization for pneumonia or bronchiolitis every year (http://www.who.int). The elderly also are at risk for severe RSV disease and 14 000 to 62 000 RSV-associated hospitalizations of the elderly occur annually in the USA. The fight against RSV is considered by the WHO as a priority. In 2005, it was estimated that 66 000–199 000 children younger than 5 years died from RSV associated acute lower respiratory infection, with 99% of these deaths occurring in developing countries. Worldwide, RSV is associated with an annual mortality rate of 160,000-600,000 deaths. RSV has been the focus of numerous efforts to develop a vaccine. To date, there is no safe and effective vaccine available, treatment options against RSV infection is limited. The goal of the Mouse Immunology-2 experiment is to discover what triggers and leads to an increased susceptibility to an infection. These findings can be used to help treat and prevent future astronauts from getting sick, as well as protect people with more vulnerable immune systems here on Earth, such as the elderly or young children. Mouse models of viral infection provide a well established experimental system to study the immune response to RSV infection. The experiments proposed in the Mouse Immunology-2 project have not been previously performed--specifically, studies of RSV infection in mice after a space flight and comparison with control mice at ground conditions. The overarching hypothesis is that combinations of several factors, from microgravity to stress, affect the human immune system during space flight, resulting in an impaired antiviral response in the respiratory tract. Three specific hypotheses that will be tested are: 1. Space flight or ground conditions such as antiorthostatic suspension leads to an altered host response against respiratory syncytial virus (RSV); 2. This “deficient” antiviral response is characterized by a defect in both innate immunity, including interferon (IFN) type I production in the respiratory tract, and T cell-mediated immune response; 3. The central events that cause this defect in antiviral immunity are the impaired trafficking and function of dendritic cells (DC) in the respiratory mucosa. Experiment objectives: • Specific Aim 1. To characterize RSV replication in the lung and nasal turbinates, disease severity, and histopathology of the airways. These studies will be conducted in mice in the hindlimb unloading model, and compared with the results from mice flown in the Space Shuttle Discovery that were infected with RSV immediately after return to Earth. • Specific Aim 2. To determine the profile of IFN-a/ß and innate cytokine response to viral infection in bronchoalveolar lavage (BAL) and nasal tissue. These studies will be conducted in mice as in Aim1. RESEARCH DESCRIPTION Understanding the function of the immune system during space flight may have great relevance to our understanding of the process of aging and/or stress-related immune-modulation (adjustments in the level of an immune response) on Earth. In particular, studying the innate host response against pathogens during/after space flight will provide novel data on the function of the respiratory mucosal response to viral pathogens. Respiratory syncytial virus (RSV) is considered the leading cause of lower respiratory tract infection in infants and children worldwide, and may cause severe and even fatal lung disease in immunocompromised and elderly persons. Approximately two-thirds of infants are infected with RSV during the first year of life, and 90% have been infected one or more times by 2 years of age. However, both infants and adults are susceptible to recurrent infections, and accumulating evidence suggests a strong link between RSV bronchiolitis and later development of asthma-like syndromes. Previous reports show that the intensity and duration of the immune response to RSV may be determined by the local cellular and cytokine response. In models of RSV disease, mice with different genetic backgrounds respond to RSV with strain-specific cytokines and patho-physiological responses. The development of an appropriate immune response during viral infections relies on early events in innate immune cells to shape the immediate environment and dictate the course of infection. As a result of the limited opportunities for space flight, animal models have been used to study changes in immune responses. We have developed a similar approach in the hindlimb unloading model (a well-accepted ground-based

model used to simulate some of the conditions of space flight and reproduce its effects on the immune system). It has been well documented in several studies that many immunologic parameters are altered in experimental animals and human subjects who have flown in space. However, it is not fully known whether these immunologic changes could result in increased susceptibility to infection. This model mimics the bone and muscle disuse as well as the fluid shifts that occur to the head and upper torso during space flight. Hindlimb unloaded rodents also exhibit several changes in immune parameters, including decreases in interleukin and interferon production, reduction in red cell mass and plasma volume, suppressed erythropoesis, splenic atrophy. Although several components of the immune system appear to be affected by hindlimb unloading, there have been limited investigations into how these alterations affect the ability to mount an immune response to a pathogen. Several studies have been performed using systemically administered pathogens. Opportunities to perform actual space flight immune response studies are expensive and rare, making ground-based models of space flight necessary to answer many preliminary questions about the possible effects of microgravity on the immune system. Although all models of simulated space flight here on Earth have significant limitations, they can provide valuable insight to possible changes that might occur during space flight and long-term space residence.

The purpose of this study is to discover what triggers and leads to an increased susceptibility to an infection. In the present study, we evaluated these parameters for hindlimb unloading model of BALB/c mice infected with RSV. A comparison was performed between hindlimb model and our findings from the space flight experiment. For the flight experiment BALB/c were flown for 12.8-days on Space Shuttle Discovery, Space Transport System (STS)-133, and infected with RSV upon return to Earth. We hypothesized that hindlimb unloading model and space flight are responsible for altering the immune response following RSV infection in a BALB/c model.

HINDLIMB UNLOADING EXPERIMENTS AT UNIVERSITY OF TEXAS MEDICAL BRANCH IN THE PI LABORATORY

CONCLUSIONS

Space flight is a unique stress model impacted consistently or intermittently by physical stress, high G forces at the time of launch and landing, increased radiation, sleep deprivation, microgravity, and nutritional factors. Although the absence of gravity cannot accurately be simulated on Earth, hindlimb unloading (HU) mimics many of the physiological alterations caused by actual space flight. Since space flight experimental opportunities for animal studies have been so limited, infection studies have not been carried out in space. Therefore, the hindlimb unloading model has proven to be invaluable for carrying out experiments on the effects of space flight conditions on resistance to infection. The main focus of this study has been to establish and characterize the effect of hindlimb unloading model on innate immune response to RSV infection in BALB/c mouse model. The effect of hindlimb unloading model followed by RSV infection on the clinical disease parameters, viral replication, lung inflammation, BAL cellular infiltrate, production of cytokines, chemokines, and type I IFN were explored herein. Our study could yield extensive information on the mechanism of the effect of HU model followed by RSV infection on immune response.

RSV pathogenesis is a complex process dependent on the interaction of viral and host determinants. These studies are critical for developing new therapeutic strategies against RSV infections. Hindlimb unloading (HU) in rodents is a ground-based model used to simulate some of the conditions of space flight. This study was aimed to compare the effect of HU and space flight on immune response against a common viral pathogen, Respiratory Syncytial Virus (RSV), using an experimental BALB/c mouse model of infection. For ground-based HU studies, mice were suspended for 7 days prior to RSV inoculation, and after infection for duration of the experimental period or kept in regular cages (control group). For the space flight experiments, mice were assigned to one of three groups: space flight (FLT), Animal Enclosure Module (AEM) and vivarium (VIV). FLT animals flew on the Space Shuttle Discovery (STS-133) and were subjected to 12.8 days of micro-gravity. All groups were infected intranasally with RSV Long strain (5X106 pfu), or sham saline. There were no significant differences in average daily food consumptions and water intake in the HU mice. However, water intake was greater for FLT and AEM mice compared with VIV group. Following RSV infection, mice in the HU group had more severe clinical disease, slightly higher lung viral titer, and significantly higher concentration of IFN-a, MCP-1, and KC in bronchoalveolar lavage fluid compared to the control group. Similar findings were observed in RSV-infected FLT mice (compared to VIV group). Furthermore, we found that RSV infection in HU and FLT mice caused a systemic response that was characterized by significantly increased levels of serum cytokines, IL-6, IL-13, IL-17, GM-CSF, MCP-1, MIP-1ß, and TNF-a, compared to control and VIV mice. Overall, these findings suggest that HU is a useful ground-based model for evaluating some inflammatory and immune parameters that may be altered by space flight.

The data reported in this study are the results of a rare opportunity, in which we have been able to show in the BALB/c model that short-term space flight and HU suspension followed by RSV infection induced significant changes in parameters that can influence immune defense mechanisms, changes that may become significant during long-term space missions. Most importantly, animals models have proved invaluable for caring out infectious disease experiments and have increased our understanding of the biological significance of the effect of space flight conditions on the immune system. Understanding the effect of space flight on the immune response is essential to ensure the health, safety, and performance of space station crew members.

1. There were no significant differences in average daily food consumptions and water intake in the HU and VIV mice. However for the space flight experiment, water intake was greater for FLT and AEM mice compared with VIV group.

2. In our study of HU model, infection of BALB/c mice by intranasal inoculation with RSV produced observable symptoms of disease such as weight loss, and ruffled fur for all groups HU, and VIV. RSV infection was associated with similar degree of body weight loss in both groups, but disease severity score was higher in the HU group. The results from body mass for mock infected groups (VIV, and HU) shows no significant differences during the experimental period. Similar results were observed for the space flight experiment where RSV infection was associated with similar reduction in body weight for all groups, but disease severity was higher for the flight animals. In conclusion, disease severity was higher for HU model and space flight animals followed by RSV infection.

3. Consistent with increased RSV infection, the cellular population in BAL fluid was altered. RSV infection induces lung recruitment of neutrophils, which are higher for the HU group at day 1 post-infection compared with VIV group. We report similar findings for the flight animals infected with RSV compared with the ground control groups.

Task Progress:

4. In our study cultures of homogenized lungs demonstrated the presence of RSV at day 5 post-infection for all groups

HU, and VIV. In addition, there was a trend suggesting an increase in RSV loads, which was more prominent in mice from the HU group compared with VIV group. Similar results were observed for the space flight experiment, where increase in RSV loads, were observed in mice from the flight group compared with ground control groups. In conclusion, a trend suggesting increase in RSV loads at day 5 post-infection was observed for HU and space flight animals infected with RSV.

5. In this study, mouse lung tissue was investigated to evaluate a potentially health effect of HU suspension and viral infection. We found that examination of the lungs of RSV infected mice from HU group at day 5 post-infection revealed multilayered perivascular and peribronchial inflammatory infiltrates, with monocytes and occasional neutrophils invading the vascular bed, and associated mild focal necrosis of vascular endothelial cells,moderate to focally severe patchy alveolitis. The lung inflammatory score grades for alveolitis, bronchiolotis, necrosis, and interstitial penumonitis were higher for HU RSV infected mice compared with the VIV group. In contrast, no histopathological changes were observed in mock inoculated mice from HU and VIV groups. Compared with the flight experiment the pathological changes in the lungs at day 7 post-infection appeared slightly milder in animal enclosure module group compared with VIV and flight infected ones.

6. In recent years it has become clear that cytokines and chemokines involved in host response to RSV infection are thought to play a central role in the pathogenesis. The profile, relative abundance and kinetics of cytokines that are produced in the airways in the course of viral respiratory infections are all factors that contribute to shaping host immunity, inflammatory response, and ultimately viral shedding. As previously reported by others and us, space flight has been shown to alter many immunological parameters, among these affected are the production of cytokines. Following RSV infection, mice in the HU group had significantly higher concentration of IFN-a, MCP-1, and KC in bronchoalveolar lavage fluid compared to the control group. Similar findings were observed for the space flight and viral infection with significant elevation of the inducible chemoattractant protein, MCP-1 at day 1 post-infection ($p<0.01$ compared with VIV infected ones, and $p<0.005$ for AEM group). Furthermore, we found that RSV infection in HU and FLT mice caused a systemic response that was characterized by significantly increased levels of serum cytokines, IL-6, IL-13, IL-17, GM-CSF, MCP-1, MIP-1ß, and TNF-a, compared to control and VIV mice. This suggests that there are selective effects of space flight and hindlimb suspension model on immune responses, and not all cytokines and chemokines are affected in the same way.

7. Type I IFNs, among the first anti-viral agents, were discovered because of their anti-viral activity against influenza virus and seen to be central in the early anti-viral response. The type I IFNs synthesis during RSV infection may be associated closely with severity of the disease. As previously reported by others and us, RSV can induce high levels of type I IFNs immediately upon infection, but this burst was short-lived with a sharp decline in type I IFNs production as the infection progressed.

Our study provides evidence that in RSV-infected animals from HU and VIV groups, IFN-a production peaked at day 1 post-infection, but type I IFN activity decreased sharply bellow limit of detection at day 5 post-infection. We have observed similar results for the flight experiment where IFN-a production peaked at 18h post-infection for RSV-infected animals from ground control, and flight groups.

Altogether, these data demonstrate that HU model and space flight can induce high levels of type I IFNs immediately upon RSV infection, but there is a rapid decline in IFN-a production for the next time point of experiment.

ACCOMPLISHMENTS POST-FLIGHT EXPERIMENTS

We made considerable progress towards fulfilling our primary aims, as previously described above.

The following posters were presented, and 2 manuscripts will be submitted for publication:

1. “EFFECT OF SPACE FLIGHT ON INNATE IMMUNITY TO RESPIRATORY VIRAL INFECTIONS I: CLINICAL DISEASE PARAMETERS” - manuscript to be submitted to American Journal of Physiology - Lung Cellular and Molecular Physiology. Teodora Ivanciuc,1 Deepthi Kolli,1 Paula Dumars,2 Elena Sbrana,3 Yashoda Hosakote-Madaiah,1 Zinzi Blell,4 Lan Pang,1 Antonella Casola,1,4 Nicole Rayl,5 , and Roberto P. Garofalo,1,4*. 1 Division of Clinical and Experimental Immunology & Infectious Diseases (CEIID), Dept. of Pediatrics, UTMB Galveston, 2 Lockheed Martin Engineering & Science Solutions, NASA Ames Research Center , 3 Dept. of Microbiology, Dept. of Pathology - Autopsy Service, UTMB Galveston, 4 Dept. of Microbiology and Immunology, UTMB Galveston, 5 NASA Ames Research Center

2. “EFFECT OF SPACE FLIGHT ON INNATE IMMUNITY TO RESPIRATORY VIRAL INFECTIONS II: CYTOKINE and BLOOD PARAMETERS ” - manuscript to be submitted to Journal of Applied Physiology. Teodora Ivanciuc,1 Deepthi Kolli,1 Paula Dumars,2 Yashoda Hosakote-Madaiah,1 Zinzi Blell,4 Lan Pang,1 Antonella Casola,1,3 Nicole Rayl,4 , and Roberto P. Garofalo,1,3*. 1 Division of Clinical and Experimental Immunology & Infectious Diseases (CEIID), Dept. of Pediatrics, UTMB Galveston, 2 Lockheed Martin Engineering & Science Solutions, NASA Ames Research Center , 3 Dept. of Microbiology and Immunology, UTMB Galveston, 4 NASA Ames Research Center

3. “EFFECT OF SPACE FLIGHT ON INNATE IMMUNITY TO RESPIRATORY VIRAL INFECTIONS” - 8th Annual 2012 Respiratory Syncytial Virus Symposia”, Santa Fe, New Mexico. T. Ivanciuc1, D. Kolli1, P. Dumars2, E. Sbrana3, Y. Hosakote-Madaiah1, Z. Blell4, L. Pang1, A. Casola 1,4, N. Rayl 5, and R.P. Garofalo 1,4*. 1 Division of Clinical and Experimental Immunology & Infectious Diseases (CEIID), Dept. of Pediatrics, UTMB Galveston, 2 Lockheed Martin Engineering & Science Solutions, NASA Ames Research Center , 3Dept. of Microbiology, Dept. of Pathology - Autopsy Service, UTMB Galveston, 4 Dept. of Microbiology and Immunology, UTMB Galveston, 5 NASA Ames Research Center.

4. “HINDLIMB UNLOADING IN MICE AS GROUND-BASED MODEL OF SPACE FLIGHT TO STUDY IMMUNITY TO RESPIRATORY VIRAL INFECTIONS” - McLaughlin Colloquium, April 12, 2013, UTMB, Galveston. T. Ivanciuc1, D. Kolli1, E. Sbrana2, A. Casola 1,2, and R.P. Garofalo 1,2*. 1 Depts. of Pediatrics, 2Microbiology and Immunology.3 Dept. of Pathology - Autopsy Service, UTMB Galveston.

FUTURE GOALS

In this study we aim to gain detailed insight into the molecular pathogenesis of respiratory virus infections by studying the global gene expression profiling in lung and thymus from RSV infected mice from HU model and space flight

experiment. The lung is sensitive to gravity, and it is of interest to know how its function is altered in the of space. NASA has reported pulmonary abnormalities in astronauts on space missions, but the molecular changes in lung tissue remain unknown. Studies on NASA Space labs during the last 4 years have provided the first comprehensive data on the extensive changes in pulmonary function that occur in sustained microgravity. The study of gene expression profiling of cells and tissue has become a major tool for discovery. Microarray experiments allow description of genome-wide expression changes in health and disease. Microarrays are able to simultaneously monitor the expression levels of thousands of genes. Such gene expression information can be used for comparing clinically relevant groups, uncovering new subclasses of diseases, and predicting clinically important outcomes, such as the response to therapy and survival.

To gain more insight into the early pathways of the pathogenesis and immune response to RSV altered by space flight and hindlimb unloading model, we will use microarrays technology to investigate the gene expression profile in the lungs and thymus.

Future experiment objectives

Objective 1: We will investigate the microarray gene expression profiles from lung and thymus, in young adult BALB/c mice at 13 weeks of age after exposure to the space flight aboard the shuttle Discovery, STS-133 for a period of 12.8 days, and infected with respiratory syncytial virus (RSV) or phosphate buffer saline (PBS) upon return to Earth. Tissue specimens (lung and thymus) were collected at different time post-infection (day 1, 5, and 7) from space flown mice (FLT) as well as ground control mice from the animal enclosure modules (AEM) and normal cages (Vivarium, VIV). RNA from tissue samples will be extracted and an automated array analysis for gene expression will be performed. To our knowledge, our study will be the first to focus on the immune response to RSV infection in lungs and thymus of BALB/c mice exposed to space flight using microarrays technology to investigate the gene expression profiles. These data will achieve to explain, in part, the varying effects that space flight has on the expression of genes associated with general stress responses as well as specific immunological functions after RSV exposure. Our study could yield extensive information on the mechanism of the effect of space flight followed by RSV infection on immune response. This type of study could give us information that would be useful for developing countermeasures to help treat and prevent future astronauts from getting sick, as well as protect people with more vulnerable immune systems here on Earth, such as the elderly or young children.

Objective 2: We will investigate the effects of space flight on strain-dependent gene expression profiles in lung and thymus samples collected from C57Bl/6 mice flown aboard Space Shuttle Atlantis, STS-135, and BALB/c mice flown on Discovery, STS-133. Analyses will be completed to identify the set of lung genes that differed in expression between C57Bl/6 and BALB/c mice. The strain differences in gene expression in mice will be identified by comparing the chips data from C57Bl/6 mice to the data set from BALB/C mice from flight groups. In a second analysis, C57Bl/6 mice and BALB/C pulmonary gene expression levels will be evaluated for the ground control groups (the data chips representing the expression of C57Bl/6 control lung tissue will be compared with those of BALB/c control ones). Further study of these genes may yield specific candidates for subsequent clinical investigation for factors influencing pulmonary response to the space flight condition. In this study, mouse lung tissue will be investigated to evaluate a potentially serious health hazard of space flight, since the lungs are very vulnerable to increased levels of radiation, inhalation of bacteria, and low ambient oxygen.

Objective 3: We will investigate the effect of hindlimb unloading model on gene expression profiles in lung and thymus samples from adult BALB/c mice infected with RSV or saline. Analyses will be completed to identify the set of genes that differed in expression between HU and VIV groups.

Objective 4: Comparison of gene expression profiles from lung and thymus samples between space flight experiment and hindlimb unloading model will be performed.

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Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Ivanciuc T, Kolli D, Dumars P, Sbrana E, Hosakote-Madaiah Y, Blell Z, Pang L, Casola A, Rayl N, Garofalo RP. "Effect of Space Flight on Innate Immunity to Respiratory Viral Infections." 8th Annual International Respiratory Syncytial Virus Symposium, Santa Fe, New Mexico, September 27-30, 2012. 8th Annual International Respiratory Syncytial Virus Symposium, Santa Fe, New Mexico, September 27-30, 2012. , Sep-2012
Abstracts for Journals and Proceedings	Ivanciuc T, Kolli D, Sbrana E, Casola A, Garofalo RP. "Hindlimb Unloading in Mice as Ground-Based Model of Space Flight to Study Immunity to Respiratory Viral Infections." McLaughlin Colloquium, UTMB, Galveston, TX, April 12, 2013. McLaughlin Colloquium, UTMB, Galveston, TX, April 12, 2013. , Apr-2013